

Singlet-mediated masses, protected operator modules and frame interactions

A complete 49-scalar CP benchmark, exact mass-coefficient correlations and all rotated linear restrictions

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Representation-based fermion interactions, exact scalar insertion maps, Gaussian matching and finite canonical poles. A local physical CP vacuum is retained; the golden frame and global selection are not predicted.

A constructive correlated interaction

A family-singlet mediator provides a useful completion of the heavy-mass interaction. Its representation permits only an identity matrix at the heavy-quark bilinear. Combined with the existing even finite 27-dimensional mediator content, every allowed renormalizable heavy-mass scalar vertex remains a family singlet. The source and cross-sector frame interactions can remain present. This gives a complete 49-neutral-scalar benchmark with canonical six-state quark matching, a positive full scalar Hessian, nonzero physical weak CP and termwise first-order neutral-scalar determinant-phase cancellation, including both permitted singlet heavy-quark Yukawas.

The matching has predictive coefficient structure: one singlet mediator makes the two-sector matrix of nine quadratic mass-insertion coefficients rank one. All 36 two-by-two minors vanish, giving eight independent relations. An exact insertion calculation shows that scalar running preserves this rank-one condition and the absence of adjoint mass insertions for arbitrary values of all 61 source/Higgs quartics. The preserved rank-one object here is the mass-coefficient matrix; the repository's quartic exchange matrices retain their separately established running.

This construction realizes a protected correlated interaction class. Its source frames are those of an explicitly specified local joint vacuum, and its frame-potential coefficients remain inputs. It does not derive the observed CKM hierarchy, golden relation, physical 66-degree phase or the global CP-vacuum winner. The Gaussian construction preserves the base potential's competing stationary energies, including its earlier selection boundary.

The mediator representation and complete fermion endpoints

Retain the odd real source nonet and eta in each sector, the shaping signs, canonical quark metrics and the genuine 3.A6 triplets. The finite even mediator fields transform as eight-prime, nine and the real five-plus-five-prime pair. Add one even real family singlet sigma. The triplet bilinear decomposes as one plus the ordinary eight. Consequently each heavy sector has one new renormalizable Yukawa h_f sigma times the identity heavy-quark bilinear, while the finite 27 mediators have no such contraction. Both h_{up} and h_{down} are included and nonzero.

The original source fields remain odd and cannot enter a renormalizable heavy-mass vertex. The even singlet and finite mediators cannot enter a heavy-to-ordinary source vertex because that bilinear has the sector's odd shaping charge. The Higgs-heavy endpoint remains forbidden. The five original real fermion coefficients per sector, plus the two new singlet heavy Yukawas, therefore give the complete twelve-coefficient renormalizable fermion action on this field content. Its source block C_f is Hermitian and its heavy block is the real scalar $(m_f + h_f \sigma) I$. No ordinary-adjoint heavy-mass Yukawa has been omitted from an even ordinary-adjoint mediator; that mediator representation is absent from this completion.

The scalar potential may contain all symmetry-permitted singlet self interactions and contacts. The mass-phase argument below depends on the fermion vertex representations and a stable positive-mass background, rather than a restriction to Gaussian scalar self coefficients. The explicit matching example chooses a Gaussian scalar potential to make every retained tree operator and pole calculation executable. Those scalar coefficients are benchmark choices.

Exact Gaussian matching and the correlated operators

Let J_{source} be a real invariant cubic-dimension current: a linear combination of the eight source quadratics and $H^\dagger H$ with dimension-one coefficients. Let F_f denote the real heavy-quark mass bilinear, and let J_F be the sum of $h_f F_f$. The Gaussian potential contains $M^2 \sigma^2 / 2 + \sigma (J_{source} + J_F) + J_{source}^2 / (2 M^2)$. The last term is a source contact chosen to preserve the base source potential after matching.

Eliminating σ gives $\sigma = -(J_{\text{source}} + J_F)/M^2$. The matched potential is $-J_{\text{source}} J_F/M^2 - J_F^2/(2M^2)$. Thus the matched heavy mass is $(m_f - h_f J_{\text{source}}/M^2)$. The two rows of quadratic mass coefficients are $B_{fi} = -h_f g_i/M^2$. With all nine g_i and both h_f nonzero, they share one source-current vector and have rank one. In the benchmark $h_{\text{up}} = 3/50$ and $h_{\text{down}} = -1/25$, so the entire up row is minus three-halves of the down row.

Tree matching also supplies a four-fermion interaction and derivative correlations. The exact Gaussian nonlocal Lagrangian is one-half $(J_{\text{source}} + J_F)(M^2 + \Box)^{-1}(J_{\text{source}} + J_F)$, minus the source contact $J_{\text{source}}^2/(2M^2)$. Its low-momentum expansion contains $J_F^2/(2M^2)$ and positive one-half $[\partial(J_{\text{source}} + J_F)]^2/M^4$ to the fourth. Keeping the complete functional retains its correlated source metric, source/fermion derivative interactions and derivative four-fermion terms. A mass-only truncation would discard part of this matching.

At the constant vacuum these derivative terms do not create a quadratic quark kinetic correction. The full UV quark metrics are canonical, and the six-state Dirac matrices are matched directly. The induced scalar metric is separately retained through the exact full Gaussian pole calculation. The nonlocal functional is a tree matching result; it is not the complete finite one-loop EFT threshold.

A complete exact scalar insertion map

The source quadratic heavy-mass operator space has eight singlet contractions and six ordinary-adjoint contractions, giving fourteen coefficients per heavy sector. Adding $H^\dagger H$ adds one singlet, giving fifteen per sector and thirty across the two. The exact scalar-loop map is computed on all 24 real source/Higgs coordinates and all 61 quartics. Its source-only restriction covers all 52 source quartics.

The insertion beta functional is $16\pi^2 \beta(B) = \text{Tr}[(G^{-1} \text{Hess } B)(G^{-1} \text{Hess } V_4)]$. The implementation uses exact rational and $Q(\sqrt{5})$ coefficients, with the same rational kinetic metric and certified finite-group projectors as the exact quartic algebra. Every resulting quadratic polynomial is expanded coefficient by coefficient. All eight ordinary-adjoint components are checked and agree. The source map has 104 nonzero coefficients; the source/Higgs extension has 121. The driver also rechecks all 1,891 products of the full quartic table against exact tensors.

The nine singlet operators form an invariant module, and no singlet-to-adjoint coefficient occurs in the complete scalar map. This holds for every quartic coefficient vector, so it persists when those quartics themselves run. The two heavy sectors evolve through the same linear insertion matrix. If their mass-coefficient matrix is initially $B = h g^T$, its derivative has the same factorized rows with the common evolving g vector. Every two-by-two minor and its beta derivative therefore vanishes. The benchmark verifies all 36 matching minors, all 36 derivative identities and the exact row ratio in its scalar counterterm.

This is a scalar-loop closure theorem and a representation-derived tree matching relation. Gauge, fermion, mediator and derivative EFT diagrams have not been assembled into a complete anomalous-dimension matrix. Higher-dimensional adjoint operators are permitted by the low-energy symmetry; finite matching can correlate them with other operators. The complete UV neutral-scalar first-order phase calculation is supplied directly below, while full finite EFT coefficient matching is a separate calculation.

Why the full neutral-scalar first-order phase vanishes

At any background with positive heavy mass $m_{f+h_f \sigma}$, diagonalize the Hermitian C_f . Each family has the real block $[[a_f, 0], [h_i, m_{f+h_f \sigma}]]$, so its left and right two-state singular vectors are real. A real scalar mass eigenmode combines a Hermitian source variation, a real family-singlet heavy-mass vertex and the real family-diagonal Higgs vertex. The singlet heavy vertex is diagonal in the source family frame. Every off-family mass-basis vertex is therefore a real coefficient times one Hermitian source entry, and the reverse pair is a real coefficient times its conjugate. Each paired product is real; diagonal vertices are real as well.

The result holds for arbitrary real scalar mixing and real spectral weights, so both the finite B0 first-order determinant phase and its divergent coefficient vanish. Scalar tadpole shifts preserve Hermitian C_f and a real singlet heavy mass. The theorem includes the added singlet's allowed couplings without setting either to zero. It retains its first-order scope; the earlier higher determinant-phase examples do not become all-loop cancellations through this extension.

Local frame selection and canonical finite matching

The worked completion uses the retained complete source/Higgs potential, its twenty source coordinates and one physical Higgs coordinate. The finite 27 mediator currents are taken from the existing certified mediation plan. Add the singlet current with coefficients $(0.10, -0.12, 0.07, 0.08, 0.05, 0.09, -0.06, 0.11, 0.04)$, where the last coefficient multiplies $H^\dagger H$. All coefficients are nonzero. The common mediator mass is 10. Each source contact and Gaussian square is retained, preserving the joint stationary point and source Schur complement exactly.

The full physical neutral scalar count is 49, or 52 real fields before removing the three Higgs Goldstones. Its smallest full Hessian eigenvalue is approximately 1.12584×10^{-5} . The source Schur complement reproduces the retained 21-field Hessian. The pole calculation eliminates all 28 mediators at finite momentum; its 21 light roots agree with the full 49-state matrix to better than 4×10^{-14} . This retains canonical scalar pole residues alongside the quark matching.

The bare quark heavy masses are 1 and 0.9. After the singlet vacuum shift they are 0.999798423482 and 0.900134384345. Both sectors have three light states below 0.04 and three heavy states above 0.9. Exact six-state canonical matching gives physical quartet $J = 0.000262046614742$. The explicitly checked CP partner has the same six masses and charged-current magnitudes with the opposite quartet sign. This is physical weak CP in the retained local vacuum, not a phase convention.

With all 49 neutral scalar modes and both singlet heavy Yukawas included, the up/down first-order neutral phases have magnitude below 4×10^{-17} . The relative correction norms are approximately 0.0327 and 0.0104. Termwise paired reality supplies the exact reason for the cancellation. These benchmark values are conditional on the declared source potential and fermion coefficients; they are not an observed-flavor fit.

The correlation test for an adjoint mass mechanism

The same insertion map sharpens the alternative in which the heavy mass has a nonzero ordinary-adjoint quadratic source component. Write the three current labels as η_A , s_A and $q(A)$, and their norm metric as $N = \text{diag}(2, 2, 10/3)$. Let C be the three-by-three matrix of cross-sector current quartics, and let $k_{\text{up}}, k_{\text{down}}$ be the two nonzero same-sector adjoint insertion vectors. Scalar generation of opposite-sector adjoints is $C^{\text{transpose}} N k_{\text{up}}$ and $C N k_{\text{down}}$. Their simultaneous vanishing gives five independent constraints and leaves four dimensions in C . The complete map verifies every opposite-sector transition, including exact zeros for the other quartics.

This one-scale kernel is not automatically preserved by the joint quartic and insertion flow. The pure-adjoint nine-operator algebra gives a stronger complete result: every real rotated linear scalar-loop-closed subspace containing both independent radial quartics and having the ordinary cross channel identically zero must also have all three finite cross channels zero. This extends the earlier coordinate-support census to all rotated real linear restrictions under these hypotheses.

For finite cross coefficients x, y, z , ordinary-channel regeneration is $Q = 6(96x^2 + 216xy + 40xz + 9y^2 + 270yz + 125z^2)/125$. Its exact signature is two positive and one negative direction. Multiplication by the sum of the two radial basis elements has eigenvalues 16, 48, 80, 128. The exact polynomial projector onto eigenvalue 16 extracts pure mixed nonsinglet components. In a closed protected subspace their finite part is totally isotropic for Q , so has dimension at most one. Closure would require a nonzero null ray whose finite-channel product stays parallel to that ray.

Explicit rational Bezout identities exclude every projective patch of that ray equation: certificate degree six for x nonzero, degree three for x zero and y nonzero, and nonzero regeneration 6 on the remaining z axis. Each identity is checked by exact polynomial expansion and included in the receipt. The proof excludes every rotated real linear restriction in the stated pure-adjoint class. Nonlinear RG-invariant varieties, restrictions correlating the radial terms, extra singlet/mediator interactions and enlarged gauge/Yukawa completions retain their distinct scopes.

Reproduction and the mechanism frontier

Run `PYTHONPATH=python OPENBLAS_NUM_THREADS=1 python python/develop_flavor_singlet_mediation.py`. The scientific receipt contains both exact insertion maps, the complete projector and quartic checks, rank-one matching and running identities, the 49-field UV benchmark, finite pole matching, the five-condition kernel and explicit polynomial obstruction certificates. Tests independently compare the operator maps with canonical Hessian contractions, check the Gaussian fermion matching, confirm the physical CP partner and exercise arbitrary real singlet/source scalar mixing. The validation receipt records focused and adjacent tests, a fresh extraction and byte-identical scientific replay.

The completed interaction demonstrates a scalar-stable correlation between source labels, cross-sector mass coefficients and quark couplings through a genuine singlet contraction, with locally selected source frames and canonical matching. The remaining predictive requirement is to constrain the frame-potential coefficients and select a global physical vacuum. Complete finite EFT matching and higher-loop CP remain physical calculations. No golden relation is asserted by the mass-module relations alone.